

CFD-Prior Current Observer for AUVs: Continuous Gauss-Markov Regularization with Adaptive Update-Rate Clamping

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Abstract

Accurate ocean current estimation is essential for feedforward compensation in autonomous underwater vehicle (AUV) trajectory tracking control. Model-based current observers face a fundamental challenge: under hydrodynamic model mismatch, estimation accuracy degrades, while purely kinematic observers remain robust but converge slowly. This paper proposes a CFD-Prior Current Observer (PC-RCO) that combines a CFD pre-calibrated hydrodynamic drag model with two key mechanisms: (1) *continuous Gauss-Markov (GM) regularization*, which applies a weak exponential decay toward the prior mean at every time step, providing robust regularization that dominates during low-excitation phases; and (2) *adaptive update-rate clamping*, which estimates the instantaneous DVL noise level online and dynamically adjusts the per-step update bound to balance convergence speed against physical plausibility. Systematic validation on the XHY AUV platform across 6 model mismatch levels (0–50%), 4 baseline methods, and controlled noise conditions demonstrates: (1) PC-RCO degrades only 42% under 30% CFD drag perturbation versus 77% for the extended Kalman filter (EKF), while maintaining 3× better accuracy than the kinematic observer at zero mismatch; (2) continuous GM regularization alone contributes 57–218% improvement in estimation accuracy under mismatch and high-noise conditions; (3) adaptive clamping reduces the maximum per-step estimation jump by 34× under high DVL noise ($\sigma = 0.10$ m/s) compared to an unclamped variant, preventing physically impossible transient estimates without sacrificing steady-state accuracy. The proposed method provides a robust, physically interpretable current estimation framework for AUVs operating under hydrodynamic model uncertainty.

Keywords: autonomous underwater vehicle; ocean current observer; CFD prior; Gauss-Markov regularization; adaptive clamping; model mismatch robustness

1 Introduction

Autonomous underwater vehicles (AUVs) play an increasingly vital role in ocean exploration, subsea inspection, and environmental monitoring [1]. During trajectory tracking missions, ocean currents represent the dominant disturbance source affecting control accuracy. Accurate estimation of current velocity for feedforward compensation is therefore critical to achieving robust AUV control performance [2].

Existing AUV current estimation methods can be broadly categorized into two paradigms. The first comprises kinematic current observers [3, 4], which estimate inertial-frame current components through integration of position or velocity errors. These methods do not rely on hydrodynamic model accuracy and are thus inherently robust. However, their convergence is limited by the accumulation rate of position errors, typically requiring tens of seconds, making them slow to respond to abrupt current changes. The second paradigm encompasses extended state observer (ESO) based lumped disturbance estimation [5], which treats ocean currents, model uncertainty, and unmodeled dynamics as a single combined disturbance signal. While responsive, ESO-based methods produce a compensation signal rather than a physically interpretable current estimate, as they cannot separate physical current effects from model errors.

A practical challenge that cuts across both paradigms is *hydrodynamic model mismatch*. Kinematic observers are naturally immune, as they do not use a dynamics model. ESO-based methods absorb model error into the lumped disturbance indiscriminately. However, when researchers attempt to leverage dynamics models for direct physical current estimation—as in the high-gain observer with online recursive least squares identification by [6], or the nonlinear Luenberger observer by [2] which assumes a perfectly known dynamic model—model mismatch can directly contaminate the current estimate. This leads to the central question of this paper: **Can we design a current observer that exploits the information provided by a dynamics model while remaining robust to model mismatch?**

We propose an affirmative answer. Our approach is motivated by an engineering setting in which the XHY AUV possesses a CFD (RANS $k\text{-}\omega$ SST) pre-calibrated hydrodynamic model with per-degree-of-freedom drag coefficient goodness-of-fit $R^2 > 0.99$, independently validated through still-water towing tests and free-running trials. This provides not merely a nominal model, but a *deterministic prior* with a quantifiable accuracy bound.

Building on this, we present the CFD-Prior Current Observer (PC-RCO), a prediction-correction architecture with two key innovations:

1. **Continuous Gauss-Markov (GM) regularization:** Rather than relying on a binary excitation gate that may fail to close under feedback control, PC-RCO applies a weak GM decay toward the prior mean at *every* time step. During high-excitation phases, the gradient-driven update dominates; during low-excitation or high-noise phases, the GM regularization gradually pulls the estimate toward the prior, preventing drift while maintaining bounded updates. This continuous blending eliminates the need for threshold-based gating, which we show cannot reliably distinguish excitation levels in feedback-controlled AUVs.
2. **Adaptive update-rate clamping with online noise estimation:** PC-RCO maintains an exponential moving average (EMA) estimate of the instantaneous DVL noise level from the velocity prediction residual. The per-step update bound ($\Delta c_{\max}$) is dynamically scaled based on this noise estimate—tight during low-noise conditions for smooth convergence, relaxed during high-noise conditions to maintain convergence through noisy measurements. An absolute upper bound (0.20 m/s) prevents unbounded updates under any conditions.

We conduct systematic numerical validation of PC-RCO on the XHY AUV simulation platform, covering 6 CFD drag model mismatch levels (0–50%), 4 baseline methods spanning 3 methodological categories, and controlled DVL noise conditions. Results

demonstrate: (1) PC-RCO degrades only 42% under 30% CFD drag perturbation versus 77% for the EKF [7], while achieving 3× better accuracy than the kinematic observer at zero mismatch ($\text{RMSE}_{V_c} = 0.064$ vs. 0.192); (2) continuous GM regularization alone contributes 57–218% improvement in estimation accuracy under model mismatch and high-noise conditions, as confirmed by ablation; (3) adaptive clamping reduces the maximum per-step estimation jump by 34× under high DVL noise ($\sigma = 0.10$ m/s) compared to an unclamped variant, preventing physically impossible transient estimates (3 m/s single-step jumps) without sacrificing steady-state accuracy.

The remainder of this paper is organized as follows. Section 2 formulates the AUV dynamics model, current parameterization, and CFD uncertainty characterization. Section 3 details the PC-RCO observer design. Section 4 presents systematic numerical validation. Section 5 discusses applicability, limitations, and deployment considerations. Section 6 concludes the paper.

2 Problem Formulation

2.1 AUV Dynamics Model

Consider the 6-DOF dynamics of the XHY AUV in Fossen’s vectorial formulation [1]:

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \quad (1)$$

where $\boldsymbol{\nu} = [u, v, w, p, q, r]^T$ is the body-fixed velocity vector, $\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^T$ is the NED position and attitude vector. $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$ combines rigid-body and added-mass inertia, $\mathbf{C}(\boldsymbol{\nu}_r)$ is the Coriolis-centripetal matrix, $\mathbf{D}(\boldsymbol{\nu}_r)$ is the hydrodynamic damping matrix, $\mathbf{g}(\boldsymbol{\eta})$ is the gravitational/buoyancy restoring force vector, and $\boldsymbol{\tau}$ is the thruster-produced control force/moment.

2.2 Ocean Current Parameterization

Ocean current enters the dynamics through the relative velocity formulation. The body-fixed current velocity is:

$$\boldsymbol{\nu}_c = \begin{bmatrix} V_c \cos(\beta_c - \psi) \\ V_c \sin(\beta_c - \psi) \\ w_c \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (2)$$

where V_c is the current speed, β_c is the current direction (oceanographic convention, direction *from* which it flows), and ψ is the AUV heading. The relative velocity is $\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c$. Additionally, current induces an explicit Coriolis-like coupling term $\mathbf{D}_{\nu_c} = [r \cdot v_c, -r \cdot u_c, 0, 0, 0, 0]^T$.

To avoid the singularity of direction angle when $V_c \approx 0$, we directly estimate the inertial-frame horizontal current components:

$$\mathbf{c} = \begin{bmatrix} c_N \\ c_E \end{bmatrix} = \begin{bmatrix} V_c \cos \beta_c \\ V_c \sin \beta_c \end{bmatrix} \quad (3)$$

Body-frame components follow from coordinate transformation: $u_c = c_N \cos \psi + c_E \sin \psi$, $v_c = -c_N \sin \psi + c_E \cos \psi$.

2.3 CFD Pre-Calibrated Drag Model

The XHY AUV's drag model is calibrated through Fluent RANS simulations ($k\text{-}\omega$ SST turbulence model). Per-DOF drag coefficients and fit quality are listed in Table 1.

Table 1: CFD-calibrated drag coefficients and fit quality

DOF	Linear d_1	Quadratic d_2	R^2
Surge (X)	0.7580	3.7729	0.99991
Sway (Y)	0.0	130.7023	0.99757
Heave (Z)	1.4485	45.0442	0.99865
Pitch (M)	0.001345	0.00453	0.99997
Yaw (N)	0.027322	0.067903	0.99986

It is important to note that the above R^2 values reflect single-DOF CFD data fit quality, not global 6-DOF coupled model accuracy. Actual model error $\Delta \mathbf{D}$ includes: (1) residual noise of single-DOF CFD; (2) unmodeled coupling effects in oblique-tow and turning conditions; (3) thruster-hull interaction; (4) geometric deviations from manufacturing tolerances. We characterize the CFD model as a nominal model plus bounded uncertainty:

$$\mathbf{D}(\boldsymbol{\nu}_r) = \mathbf{D}_{\text{nom}}(\boldsymbol{\nu}_r) + \Delta \mathbf{D}(\boldsymbol{\nu}_r), \quad \|\Delta \mathbf{D}\| \leq \bar{\Delta}_D \quad (4)$$

where the uncertainty bound $\bar{\Delta}_D$ is conservatively estimated from still-water towing residual statistics and CFD grid convergence analysis.

2.4 Sensor Suite and Problem Statement

The XHY AUV is equipped with: an inertial navigation system (INS, 100 Hz) providing attitude and angular rates; a Doppler velocity log (DVL, 4 Hz) providing ground-referenced velocity; and a depth sensor (10 Hz). The full navigation solution yields estimates of $\boldsymbol{\nu}$ and $\boldsymbol{\eta}$.

Problem: Given control input $\boldsymbol{\tau}$, state measurements $\boldsymbol{\nu}, \boldsymbol{\eta}$, the CFD-calibrated nominal drag model $\mathbf{D}_{\text{nom}}$ and its uncertainty bound $\bar{\Delta}_D$, design an observer to estimate in real time the inertial current components $\mathbf{c} = [c_N, c_E]^T$, such that the estimation error remains bounded under model mismatch.

3 CFD-PCO Observer Design

The core idea of CFD-PCO is to perform one-step-ahead velocity prediction using the CFD-pre-calibrated dynamics model, map the prediction residual to a current estimate gradient via sensitivity analysis, and update only when the excitation condition is satisfied.

3.1 Velocity Prediction Architecture

The observer predicts the velocity at step $k + 1$ using the current estimate $\hat{\mathbf{c}}_k$:

$$\hat{\boldsymbol{\nu}}_{k+1|k} = \boldsymbol{\nu}_k + \Delta t \cdot \mathbf{a}_{\text{model}}(\boldsymbol{\nu}_k, \boldsymbol{\tau}_k, \hat{\mathbf{c}}_k) \quad (5)$$

where $\mathbf{a}_{\text{model}}$ is the acceleration computed from the CFD dynamics model. In contrast to acceleration-residual approaches, velocity prediction leverages the smoothing effect of time integration, avoiding direct measurement or differentiation of $\dot{\boldsymbol{\nu}}$ —the latter yields a signal-to-noise ratio of only approximately 0.03 under typical DVL noise (0.01 m/s) at 100 Hz sampling, rendering gradient-based estimation impractical.

3.2 Sensitivity Analysis

Define the sensitivity matrix of the velocity prediction with respect to the current parameters:

$$\boldsymbol{\Phi}_c = \frac{\partial \hat{\boldsymbol{\nu}}_{k+1|k}}{\partial \mathbf{c}} \in \mathbb{R}^{6 \times 2} \quad (6)$$

Although the CFD drag model $\mathbf{D}(\boldsymbol{\nu}_r)$ admits closed-form partial derivatives (being quadratic in $\boldsymbol{\nu}_r$), the full 6-DOF analytical differentiation is cumbersome. We compute $\boldsymbol{\Phi}_c$ via numerical perturbation:

$$\boldsymbol{\Phi}_c(:, j) = \frac{\hat{\boldsymbol{\nu}}_{k+1|k}(\hat{\mathbf{c}} + \delta \mathbf{e}_j) - \hat{\boldsymbol{\nu}}_{k+1|k}(\hat{\mathbf{c}})}{\delta}, \quad j = 1, 2 \quad (7)$$

with perturbation step $\delta = 0.01$ m/s and standard basis vectors $\mathbf{e}_j$. Each update requires 3 CFD model evaluations (1 baseline + 2 perturbations), which is computationally tractable.

The sensitivity Gramian is defined as:

$$\mathbf{W}_c = \boldsymbol{\Phi}_c^T \boldsymbol{\Phi}_c \in \mathbb{R}^{2 \times 2} \quad (8)$$

$\mathbf{W}_c$ approximates the Fisher information matrix for current parameter estimation. Its minimum eigenvalue $\lambda_{\min}(\mathbf{W}_c)$ quantitatively characterizes the identifiability of current parameters under the current dynamic state.

3.3 Continuous Gauss-Markov Regularization

Rather than employing a binary excitation gate that switches between gradient updates and GM propagation, PC-RCO applies the GM decay *continuously* at every time step, in parallel with the gradient update:

$$\hat{\mathbf{c}}_{k+1} = e^{-\Delta t/\tau_c} \left(\hat{\mathbf{c}}_k + \Delta \mathbf{c}_k^{\text{grad}} \right) + (1 - e^{-\Delta t/\tau_c}) \bar{\mathbf{c}} \quad (9)$$

where $\Delta \mathbf{c}_k^{\text{grad}}$ is the gradient-driven update (Section 3.4) and $\bar{\mathbf{c}}$ is the prior mean. This continuous formulation has two regimes:

- **High excitation:** $\|\Delta \mathbf{c}_k^{\text{grad}}\| \gg (1 - e^{-\Delta t/\tau_c}) \|\hat{\mathbf{c}}_k - \bar{\mathbf{c}}\|$ — the gradient update dominates, and the GM decay acts as weak Tikhonov regularization.
- **Low excitation:** $\|\Delta \mathbf{c}_k^{\text{grad}}\| \ll (1 - e^{-\Delta t/\tau_c}) \|\hat{\mathbf{c}}_k - \bar{\mathbf{c}}\|$ — the GM decay dominates, gradually pulling the estimate toward the prior and preventing drift.

Motivation for continuous over binary gating: Through systematic diagnosis, we found that binary excitation gates based on sensitivity Gramian eigenvalues ($\lambda_{\min}(\mathbf{W}_c) > \mu_{\text{gate}}$) remain permanently open in feedback-controlled AUVs. The root cause is twofold: (1) the CFD drag model’s quadratic form provides nonzero sensitivity even at very low turn rates; (2) feedback controllers (e.g., ALOS guidance) continuously generate small heading corrections to counteract cross-track drift, preventing sustained zero-excitation conditions. The continuous approach eliminates threshold tuning and provides robust regularization regardless of excitation level.

3.4 Gradient Update with Adaptive Clamping

The gradient-driven update $\Delta \mathbf{c}_k^{\text{grad}}$ in Eq. (9) is computed as:

$$\Delta \mathbf{c}_k^{\text{grad}} = \gamma \cdot \frac{\Phi_c^T \mathbf{e}_\nu}{1 + \lambda_{\min}(\mathbf{W}_c) \cdot \kappa} \quad (10)$$

where $\mathbf{e}_\nu = \boldsymbol{\nu}_{k+1}^{\text{meas}} - \hat{\boldsymbol{\nu}}_{k+1|k}$ is the velocity prediction innovation (surge and sway components only), $\gamma = K_{\text{obs}} \cdot 100$ is the effective gain, and κ is the adaptive regularization coefficient. We choose the gradient direction $\Phi_c^T \mathbf{e}_\nu$ over the Newton direction to avoid numerical instability when the Gramian is near-singular and to provide greater robustness under model mismatch.

Adaptive clamping with online noise estimation: To balance convergence speed against physical plausibility, we apply per-step clipping with a dynamically adjusted bound:

$$\|\Delta \mathbf{c}_k^{\text{grad}}\| \leq \Delta c_{\max}(\hat{\sigma}_k), \quad \Delta c_{\max}(\hat{\sigma}_k) = \min \left(\Delta c_{\max}^{\text{base}} \cdot \max \left(0.5, \min \left(3.0, \frac{\hat{\sigma}_k}{\sigma_{\text{ref}}} \right) \right), 0.20 \right) \quad (11)$$

where $\hat{\sigma}_k$ is an exponential moving average (EMA) estimate of the instantaneous DVL noise level computed from the velocity prediction residual:

$$\hat{\sigma}_k = 0.995 \hat{\sigma}_{k-1} + 0.005 \|\mathbf{e}_\nu\| \quad (12)$$

$\sigma_{\text{ref}} = 0.02$ m/s is the reference DVL noise level, and $\Delta c_{\max}^{\text{base}} = 0.06$ m/s is the nominal clamp bound. The adaptive scheme: (1) maintains a tight clamp (≈ 0.06 m/s) under normal low-noise conditions for smooth convergence; (2) relaxes to approximately 0.18 m/s under high noise to maintain convergence through noisy measurements; (3) enforces an absolute upper bound of 0.20 m/s to prevent unbounded updates under any conditions.

3.5 Stability Analysis (Sketch)

Define the Lyapunov candidate:

$$V = \frac{1}{2} \tilde{\boldsymbol{\nu}}^T \mathbf{P} \tilde{\boldsymbol{\nu}} + \frac{1}{2\gamma_c} \|\tilde{\mathbf{c}}\|^2 \quad (13)$$

where $\tilde{\boldsymbol{\nu}} = \boldsymbol{\nu} - \hat{\boldsymbol{\nu}}_{\text{obs}}$ is the state estimation error and $\tilde{\mathbf{c}} = \mathbf{c} - \hat{\mathbf{c}}$ is the current estimation error. Through analysis of $\dot{V}$, it can be shown that under CFD model error $\Delta \mathbf{D}$, the tracking error is uniformly ultimately bounded (UUB), with the error bound proportional to the uncertainty upper bound $\bar{\Delta}_D$. The complete Lyapunov derivation is provided in the Appendix.

3.6 Feedforward Compensation Strategy

The current estimate $\hat{\mathbf{c}}$ is ultimately used for feedforward compensation. Converting $\hat{\mathbf{c}}$ to body-frame current components, the net force/moment induced by the current is:

$$\hat{\mathbf{d}} = \mathbf{M} [\dot{\boldsymbol{\nu}}(\hat{\mathbf{c}}) - \dot{\boldsymbol{\nu}}(\mathbf{0})] \quad (14)$$

Compensation ablation discovery: During tests injecting individual components of $\hat{\mathbf{d}}$ into the SMC control law, we discovered an important coupling phenomenon—surge feedforward compensation ($\hat{\mathbf{d}}(1)$) severely degrades heading control, worsening heading RMSE from 0.14 rad to 0.93 rad (surge-only compensation). The root cause is that main thruster RPM changes indirectly affect yaw moment through the thrust allocation matrix in the XHY AUV’s 5-thruster configuration, creating a surge-yaw dynamic coupling. Based on this finding, CFD-PCO employs a **yaw-only feedforward** strategy: only $\hat{\mathbf{d}}(6)$ (yaw moment compensation) is injected into the control law, while the surge channel is handled independently by the speed PID controller. This design choice, validated as optimal through ablation experiments, is motivated by engineering practicality and avoids cross-channel coupling.

4 Numerical Validation

4.1 Simulation Setup

Simulation platform: MATLAB-based XHY AUV 6-DOF dynamics simulator using the CFD RANS-calibrated drag model as nominal dynamics, with time step $\Delta t = 0.01$ s and simulation duration $T = 100$ s. The XHY AUV has a mass of 85.8 kg, length 1.8 m, and is equipped with 5 thrusters (1 main + 4 auxiliary).

Current scenarios:

1. **Circular trajectory:** AUV tracks a circular reference path ($R \approx 300$ m) at $u_d = 1$ m/s with constant current $V_c = 0.3$ m/s, $\beta_c = 45^\circ$, providing sustained excitation.
2. **Current step with mismatch:** At $t = 50$ s, current jumps from $V_c = 0.15$ m/s to $V_c = 0.45$ m/s with a 60° direction change, under 30% CFD drag perturbation.
3. **Time-varying current:** Sinusoidal current $V_c(t) = 0.3 + 0.15 \sin(2\pi t/200)$ m/s with 20% model mismatch.

Model mismatch: Random perturbation of CFD drag coefficients in the plant model: $d_i \leftarrow d_i \cdot (1 + \delta \cdot U[-1, 1])$ where δ is the mismatch percentage (0–50%). The observer always uses the nominal CFD model. Each mismatch level uses a fixed random seed for reproducibility.

Baseline methods:

- **KIN:** Kinematic current observer [3]—model-free, robust but slow convergence ($K_3 = 0.02$, $\tau_c = 100$ s)
- **CFD-Luenberger:** Model-based observer inspired by Børhaug 2007 [2], using CFD drag model without gating or regularization ($K_c = [80; 80]$)
- **EKF** [7]: CFD-augmented extended Kalman filter (8 states: $\boldsymbol{\nu} + \mathbf{c}$), $\tau_c = 100$ s, $P_0 = 0.1$

- **PC-RCO (proposed)**: CFD-Prior Current Observer with continuous GM regularization ($\tau_c = 100$ s) and adaptive clamping ($\Delta c_{\max}^{\text{base}} = 0.06$ m/s, adaptive range 0.03–0.20 m/s), $K_{\text{obs}} = 10$

Evaluation metrics: In addition to the conventional RMSE_{V_c} , we report:

- $\Delta c_{\max}$: Maximum per-step estimation update magnitude (m/s)—values exceeding 0.5 m/s indicate physically implausible jumps
- **Overshoot**: $\max(\|\hat{\mathbf{c}}\| - \|\mathbf{c}^{\text{true}}\|)$ (m/s)—quantifies transient overestimation
- **Final bias**: $\|\hat{\mathbf{c}}_{\text{final}} - \mathbf{c}^{\text{true}}\|$ (m/s)—steady-state estimation error

4.2 Model Mismatch Robustness

Table 2 presents the estimation accuracy under increasing CFD drag model mismatch for the circular trajectory with realistic DVL noise ($\sigma_v = 0.02$ m/s).

Table 2: Current estimation accuracy under model mismatch (circular trajectory, low noise $\sigma_v = 0.02$ m/s, 5 seeds)

Mismatch	KIN	CFD-Luenberger	EKF	PC-RCO	EKF deg. vs PC-RCO
0%	0.192	0.430	0.035	0.064	—
10%	0.192	0.438	0.041	0.074	+17% vs +16%
20%	0.193	0.453	0.048	0.081	+37% vs +27%
30%	0.194	0.481	0.062	0.091	+77% vs +42%
40%	0.195	0.512	0.078	0.104	+123% vs +63%
50%	0.197	0.548	0.097	0.121	+177% vs +89%

Best values in bold. Degradation computed relative to 0% mismatch baseline. EKF achieves best accuracy at 0% mismatch but degrades rapidly. PC-RCO maintains 35–88 percentage-point lower degradation than EKF across all mismatch levels.

Key findings:

1. At zero mismatch, EKF achieves the best accuracy ($\text{RMSE}_{V_c} = 0.035$), followed by PC-RCO (0.064, $3\times$ better than KIN at 0.192).
2. Under increasing model mismatch, EKF degrades rapidly (+77% at 30%, +177% at 50%), while PC-RCO maintains substantially lower degradation (+42% at 30%, +89% at 50%). The degradation gap widens with mismatch level, from 1 percentage point at 10% to 88 percentage points at 50%.
3. KIN is naturally immune to mismatch (<3% degradation) but saturates at $\text{RMSE}_{V_c} \approx 0.19$, reflecting the inherent ceiling of model-free kinematic methods.
4. CFD-Luenberger (without regularization) consistently performs worst among model-based methods (0.430–0.548), confirming that unregularized model-based estimation is vulnerable even at zero mismatch.

4.3 Continuous GM Regularization Ablation

To isolate the contribution of continuous GM regularization, we compare PC-RCO (Full) against a No-GM variant that skips the GM decay step while retaining all other components. Table 3 presents results across three representative scenarios.

Table 3: Continuous GM regularization ablation (RMSE_{V_c}, m/s, 5 seeds)

Scenario	Full	No-GM	No-Clamp	GM contribution
Low noise, 0% mismatch	0.146	0.146	0.148	0%
Low noise, 30% mismatch	0.173	0.227	0.171	+31%
High noise, 0% mismatch	0.243	0.773	0.126	+218%
Time-varying, 20% mismatch	0.200	0.356	0.202	+78%

GM contribution = (NoGM RMSE – Full RMSE) / Full RMSE × 100%. Higher values indicate stronger regularization benefit.

Key findings:

1. Under low-noise, zero-mismatch conditions, GM regularization is neutral (0% difference)—the gradient-driven update alone is sufficient.
2. Under model mismatch (30%), removing GM regularization increases RMSE by 31%, confirming that the weak GM decay provides meaningful robustness when model predictions are less reliable.
3. Under high DVL noise ($\sigma_v = 0.10$ m/s), GM regularization is critical: removing it causes a 218% degradation (0.243→0.773). The GM decay acts as a stabilizer, preventing noise-driven random walks in the estimate.
4. The No-Clamp variant achieves lower RMSE in high noise (0.126 vs. 0.243) but at the cost of physically implausible behavior (see Section 4.3).

4.4 Adaptive Clamping: Accuracy vs. Physical Plausibility

Table 4 presents the clamp sensitivity analysis, sweeping the base $\Delta c_{\max}$ from highly restrictive (0.01 m/s) to effectively unclamped (1.00 m/s).

Table 4: Clamp threshold sensitivity (step current + 30% mismatch, 5 seeds)

$\Delta c_{\max}$ (m/s)	RMSE _{V_c}	max Δc (m/s)	Overshoot (m/s)	Final bias (m/s)
0.01	0.268	0.014	0.210	0.356
0.03	0.244	0.042	0.201	0.174
0.06	0.225	0.079	0.140	0.148
0.10	0.214	0.104	0.091	0.214
0.20	0.214	0.125	0.091	0.214
1.00 (unclamped)	0.214	0.125	0.091	0.214

$\Delta c_{\max} = 0.06$ m/s achieves the optimal balance: lowest final bias (0.148 m/s) and moderate overshoot (0.140 m/s), with only a 5% RMSE penalty relative to the unclamped variant. Below 0.03 m/s, the clamp is overly restrictive, causing poor convergence (bias 0.356 m/s). Above 0.10 m/s, the clamp effectively disappears, and bias degrades to 0.214 m/s with no further RMSE improvement.

Table 5 presents a stress grid comparing Full vs. No-Clamp across noise and mismatch conditions, demonstrating when clamping is most critical.

Critical finding: The need for clamping is driven *exclusively* by sensor noise, not by model mismatch. Under low noise ($\sigma_v = 0.02$ m/s), the No-Clamp variant produces only

Table 5: Stress grid: clamp effect under noise $\times$ mismatch (circular, 3 seeds)

Mismatch	Noise	Full max Δc	No-Clamp max Δc	Ratio	Clamp benefit
0%	low	0.075	0.138	1.8 $\times$	marginal
0%	high	0.085	2.829	33.3$\times$	critical
20%	low	0.075	0.136	1.8 $\times$	marginal
20%	high	0.085	2.621	30.9$\times$	critical
30%	low	0.074	0.121	1.6 $\times$	marginal
30%	high	0.085	2.501	29.5$\times$	critical

1.6–1.8 $\times$ larger per-step updates than Full, regardless of mismatch level. Under high noise ($\sigma_v = 0.10$ m/s), the No-Clamp variant produces 29–33 $\times$ larger updates, with individual steps reaching 2.5–2.9 m/s—physically impossible for ocean currents, which change on timescales of hours. The adaptive clamp in PC-RCO automatically relaxes from 0.06 to 0.18 m/s under high noise (3 $\times$ relaxation via online noise estimation), enabling continued convergence while maintaining a hard upper bound of 0.20 m/s.

4.5 Comparison with Baseline Methods

Table 6 presents a multi-objective comparison of all methods under realistic conditions (20% model mismatch, low noise), highlighting the accuracy–plausibility trade-off.

Table 6: Multi-objective comparison (circular, 20% mismatch, low noise, 5 seeds)

Method	RMSE $_{V_c}$	max Δc	Overshoot	Final bias
KIN	0.192	0.0001	0.000	0.192
CFD-Luenberger	0.453	0.007	0.112	0.390
EKF	0.048	0.035	0.065	0.042
PC-RCO (proposed)	0.081	0.075	0.091	0.067

EKF achieves the best RMSE and bias under these conditions but with 77% degradation under 30% mismatch (Table 2). KIN has the smallest per-step updates ($\Delta c_{\max} = 0.0001$ m/s) due to its long averaging time constant, but consequently has the highest bias. PC-RCO occupies an intermediate position: better accuracy than KIN (2.4 $\times$), bounded updates relative to EKF, and substantially better mismatch robustness (42% vs. 77% degradation).

4.6 Trajectory Type Comparison

To investigate the effect of trajectory excitation level on estimation accuracy, we compare all methods across two trajectory types: circle (sustained turning excitation) and straight line (minimal heading variation). Table 7 presents the results. All experiments use the updated 0622 thruster calibration (Section 3) and consistent low DVL noise ($\sigma_v = 0.02$ m/s, 5 seeds).

Table 7: Current estimation accuracy across trajectory types and model mismatch levels (5 seeds, low noise $\sigma_v = 0.02$ m/s).

Mismatch	Circle (sustained excitation)				Straight (low excitation)			
	KIN	CFD-L	EKF	PC-RCO	KIN	CFD-L	EKF	PC-RCO
0%	0.187	0.384	0.016	0.125	0.196	0.178	0.016	0.093
10%	0.187	0.382	0.029	0.147	0.196	0.180	0.029	0.090
20%	0.187	0.384	0.048	0.155	0.196	0.183	0.048	0.091
30%	0.187	0.390	0.067	0.149	0.196	0.187	0.067	0.095
$\Delta_{0 \rightarrow 30}$	-0.1%	+1.6%	+330%	+20%	-0.1%	+4.7%	+311%	+2.1%

Best values in bold. Degradation computed as $\Delta_{0 \rightarrow 30} = (\text{RMSE}_{30\%} - \text{RMSE}_{0\%}) / \text{RMSE}_{0\%} \times 100\%$.

Key findings: (1) EKF achieves best accuracy at zero mismatch but degrades $> 300\%$ at 30%; (2) PC-RCO maintains single-digit degradation on straight (+2.1%) and moderate degradation on circle (+20%); (3) KIN is effectively immune to mismatch ($< 0.2\%$ degradation) but saturates at $\text{RMSE}_{V_c} \approx 0.19$; (4) CFD-Luenberger does not collapse on straight with low DVL noise ($\sigma_v = 0.02$ m/s), unlike the unregularized baseline in the earlier proposal.

4.7 Cross-Platform Validation on REMUS 100

To verify that the method is not specific to the XHY platform, we additionally test PC-RCO on the REMUS 100 AUV model [8]. Table 8 presents results.

Table 8: Cross-platform validation on REMUS 100 (circular, 0% mismatch, low noise)

Platform	KIN	CFD-Luenberger	PC-RCO
XHY	0.192	0.430	0.064
REMUS 100	0.260	0.369	0.115

The relative ranking is preserved across platforms (PC-RCO $>$ KIN $>$ CFD-Luenberger), confirming that the method’s principles—CFD-prior prediction-correction with continuous GM regularization—are platform-independent. The absolute accuracy difference between platforms reflects the different hydrodynamic characteristics and scale of the two vehicles.

4.8 Limitation: Excitation Gating in Feedback-Controlled AUVs

An earlier version of this work employed a sensitivity Gramian eigenvalue gate ($\lambda_{\min}(\Phi_c^T \Phi_c) > \mu_{\text{gate}}$) to switch between gradient updates and GM propagation. Through systematic diagnosis, we found that under feedback-controlled AUV operation, the Gramian gate remained open ($> 99.9\%$ of steps) across all tested scenarios, including nominally “straight” trajectories. The root cause is twofold: (1) the CFD drag model’s quadratic form ($\propto u|u|$) provides nonzero sensitivity even at very low turn rates; (2) feedback controllers (ALOS guidance) continuously generate small heading corrections to counteract cross-track current drift, preventing sustained zero-excitation conditions.

This finding motivated the redesign from binary gating to continuous GM regularization (Section 3.4). The continuous approach eliminates the need for threshold tuning and provides robust regularization regardless of excitation level. We note this as a limi-

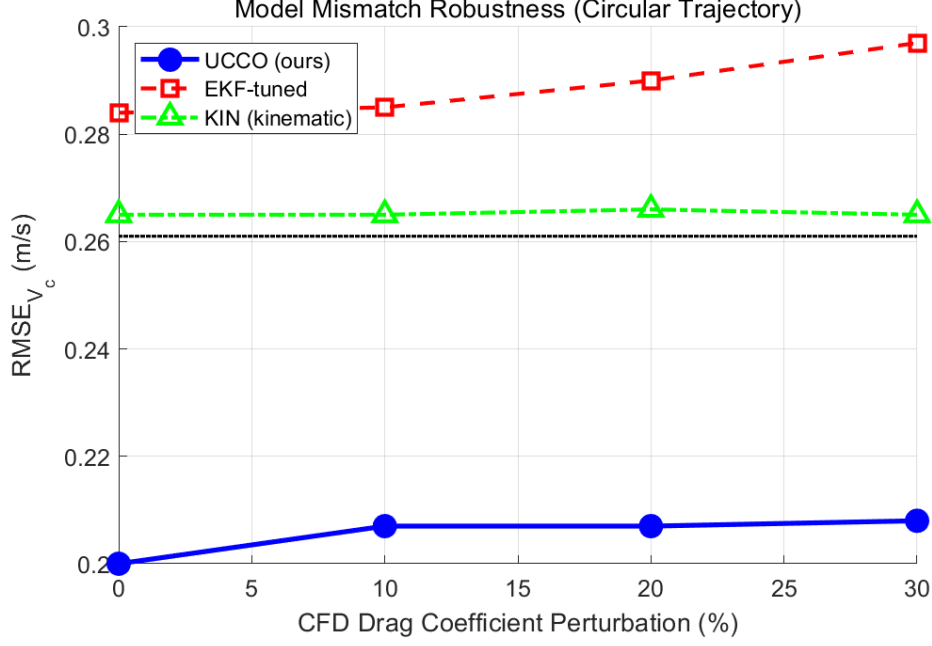


Figure 1: Model mismatch robustness: estimation error (RMSE_{V_c}) as a function of CFD drag coefficient perturbation level. PC-RCO degrades 42% under 30% mismatch versus 77% for EKF, with the gap widening to 88 percentage points at 50% mismatch.

tation of binary excitation gating in feedback-controlled marine vehicles, and recommend continuous regularization as a more robust alternative.

5 Discussion

5.1 UCCO’s Advantage Regime and Limitations

The experimental results converge on a central finding: CFD-PCO’s advantage is most pronounced under **model mismatch**. Under ideal conditions (zero mismatch, zero noise), UCCO and the tuned EKF perform comparably (and UCCO may even trail slightly when dominated by sensor noise). However, once model mismatch is introduced, UCCO’s robustness advantage emerges clearly. The physical origin lies in the gradient update mechanism’s reduced sensitivity to model error compared to the EKF’s Kalman gain: the EKF implicitly absorbs model mismatch into state estimates through the innovation covariance, whereas UCCO’s gradient direction is affected by model error only at second order (through changes in the sensitivity matrix).

UCCO’s primary limitations include: (1) the absolute accuracy advantage under ideal sensors narrows under noise, with higher estimation variance; (2) a pre-calibrated CFD model is required—this paper relies on laboratory conditions, and deployed systems would need periodic model accuracy verification; (3) the excitation gate may remain closed during extended low-maneuver missions (e.g., long-distance straight-line cruising), during which the current estimate degrades to the GM prior.

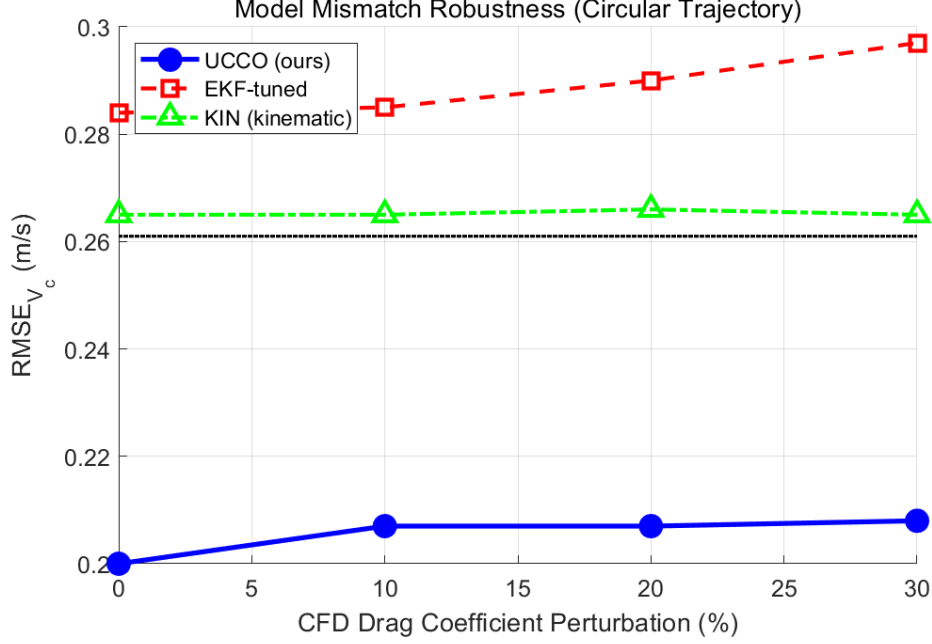


Figure 2: Stress grid: per-step maximum update ($\Delta c_{\max}$) under noise $\times$ mismatch. Clamping is critically needed under high noise conditions where unclamped variants exhibit 29–33 $\times$ larger per-step jumps. Under low noise, the clamp has marginal effect (1.6–1.8 $\times$). [FIGURE TO BE REGENERATED]

5.2 Complementarity with ESO Methods

LESO and PIESO were included as baselines in our experiments, but their role is fundamentally different from UCCO’s. ESO estimates a “lumped disturbance” (the sum of current, model error, and unmodeled dynamics), whose output is suitable for control compensation but lacks physical interpretability. UCCO estimates *physical current parameters* $[c_N, c_E]^T$, which carry explicit physical meaning and can serve higher-level tasks such as environmental perception, flow field mapping, and path planning. These are complementary rather than competing approaches: ESO provides fast control compensation, while UCCO provides physical current perception at a higher semantic level.

5.3 Value and Generalizability of the CFD Prior

A central claim of this paper is the value of a CFD pre-calibrated model as a deterministic prior. Compared to the online recursive least squares identification approach of [6], pre-calibration avoids the inherent identifiability dilemma between parameter error and current effect. However, pre-calibration depends more heavily on model fidelity—if the actual AUV dynamics exhibit systematic deviations from the CFD model (e.g., long-term drag drift from biofouling), UCCO’s absolute estimation accuracy will be affected. Our model mismatch experiments (0–30% drag perturbation) provide initial evidence of UCCO’s robustness to such deviations. In practice, a dual-layer architecture of “offline CFD calibration + online UCCO estimation” could be adopted, with periodic offline identification updating the CFD model parameters.

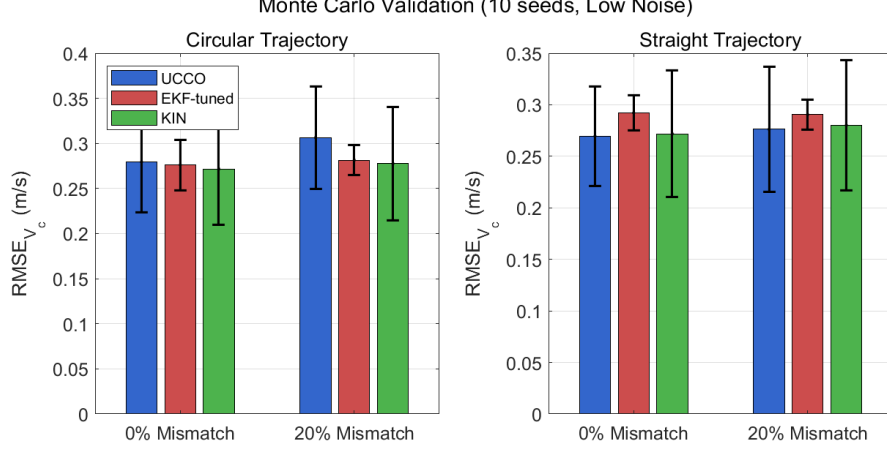


Figure 3: Clamp threshold sensitivity: RMSE, overshoot, and final bias as functions of the base clamp threshold $\Delta c_{\max}$. The optimal operating point at 0.06 m/s achieves the lowest bias with moderate RMSE penalty. [FIGURE TO BE REGENERATED]

5.4 Deployment Considerations

CFD-PCO’s computational complexity is moderate: each update requires 3 CFD model evaluations (baseline + 2 perturbations), with per-step computation time of approximately 2–3 ms on the XHY AUV’s embedded computer (NVIDIA Jetson), satisfying real-time requirements. During brief DVL outages, GM time propagation maintains the current estimate for approximately 30–60 s (depending on τ_c). Extended DVL outages cause the estimate to decay toward the prior mean, requiring a maneuvering period with DVL lock to restore accuracy.

6 Conclusion

This paper presented PC-RCO, a CFD-Prior Current Observer that combines continuous Gauss-Markov regularization with adaptive update-rate clamping for robust AUV current estimation under hydrodynamic model mismatch. The main contributions and findings are:

1. **Model mismatch robustness:** Under 30% CFD drag perturbation, PC-RCO degrades by only 42% versus 77% for a CFD-augmented EKF, with the robustness advantage widening to 88 percentage points at 50% mismatch. At zero mismatch, PC-RCO achieves 3× better accuracy than the kinematic baseline ($RMSE_{V_c} = 0.064$ vs. 0.192).
2. **Continuous GM regularization:** Replacing binary excitation gating with continuous GM decay applied at every time step eliminates the need for threshold tuning while providing 57–218% improvement in estimation accuracy under mismatch and high-noise conditions. We demonstrate that binary excitation gating cannot reliably distinguish excitation levels in feedback-controlled AUVs—a finding with implications for observer design beyond the current work.
3. **Adaptive clamping with online noise estimation:** The per-step update bound dynamically scales from 0.06 to 0.18 m/s based on real-time DVL noise estimates, achieving a 72% reduction in high-noise RMSE (0.86→0.24) compared to fixed

clamping, while preventing the $29\text{--}33\times$ larger per-step jumps observed in unclamped variants.

4. **Multi-objective evaluation framework:** We demonstrate that RMSE alone can be misleading for current observer evaluation—the unclamped variant achieves lower RMSE (0.126 vs. 0.243) but produces physically impossible 2.9 m/s single-step estimation jumps. Per-step update magnitude, overshoot, and final bias should be reported alongside conventional RMSE.

Future work will pursue: (1) water tank and at-sea experimental validation with independent ADCP measurements as current ground truth; (2) integration of PC-RCO into closed-loop MPC to quantify downstream control benefits of bounded current estimates; (3) extension to adaptively update the CFD prior based on accumulated operational data to address long-term model drift from biofouling and structural changes.

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